XUYANG LIN

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EDUCATION

New York University, New York, NY, USA **2025 – Present**

- Ph.D. student, Department of Finance and Risk Engineering
- Advisor: **Professor Nizar Touzi**

Sorbonne University, Paris, France **2024 – 2025**

- Master 2 in Probability and Random Models
- Distinction: Mention Très Bien (highest honors)

École Polytechnique, Palaiseau, France **2021 – 2024**

- French *Diplôme d'Ingénieur* in Mathematics and Applied Mathematics

Zhejiang University, Hangzhou, China **2017 - 2021**

- Bachelor in Mathematics and Applied Mathematics

PUBLICATIONS AND PREPRINTS

- Eduardo Abi Jaber, Xiaoyuan (Shaun) Li, & **Xuyang Lin** (2026). Fourier–Laplace transforms in polynomial Ornstein–Uhlenbeck volatility models. *Finance and Stochastics*, **30**(4), 1023–1079.
- Tony Lelièvre, **Xuyang Lin**, & Pierre Monmarché (2026). Convergence rates for an adaptive biasing potential scheme from a Wasserstein optimization perspective. *Nonlinearity*, **39**(4), 045016.
- **Xuyang Lin**, Mathieu Lise, & Nizar Touzi (2026). Path-Dependent Ergodic Optimal Control and Backward Stochastic Differential Equations. *arXiv preprint*, arXiv:2606.06757.
- **Xuyang Lin**, & Nizar Touzi (2026). Ergodic Linear–Quadratic Optimal Control with Stochastic Coefficients. *arXiv preprint*, arXiv:2610.30219.

TALKS AND PRESENTATIONS

The Quintic Model: A Simple Solution to the Complex SPX–VIX Calibration Problem?
Paris–Vienna Workshop, Paris, France, March 2024

Polynomial Ornstein–Uhlenbeck Volatility Models for SPX and VIX Pricing
CFA Quant Awards 2024 Award Ceremony, Amsterdam, Netherlands, April 2025

Backward SDE for Path-Dependent Ergodic Optimal Control
ORFE Seminar, Princeton University, NJ, USA, April 2026

Backward SDE for Path-Dependent Ergodic Optimal Control
Conference on Stochastic Control and Games for Risk and Regulation, Tunisia, April 2026

Path-Dependent Ergodic Optimal Control and Backward Stochastic Differential Equations
Invited Talk, School of Mathematical Sciences, Fudan University, Shanghai, China, July 2026

Path-Dependent Ergodic Optimal Control and Backward Stochastic Differential Equations
International Workshop on Probability Theory and Stochastic Analysis, Shandong University, China, August 2026

AWARDS

CFA Quant Awards Europe - the third prize **2024**

Excellence Scholarship from the École Polytechnique Foundation **2024 – 2025**

French EIFFEL Excellence Scholarship **2021 – 2024**